SMART INDIA HACKATHON 2026

AEGISNET 2.0-AI-Powered Self-Healing Environmental Monitoring Network

- **Problem Statement ID – 26178**
- **Problem Statement Title- Resilient AI-Powered Environmental Monitoring Network for Early Detection and Localized Alerts**
- **Theme- Disaster Management**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name - TerraSentinels**



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AEGISNET 2.0

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The Problem

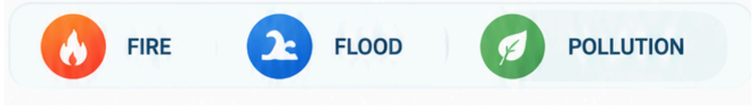
Existing monitoring (NDMA/IMD/ISRO) is centralized and loses coverage exactly when a disaster damages the infrastructure it depends on.

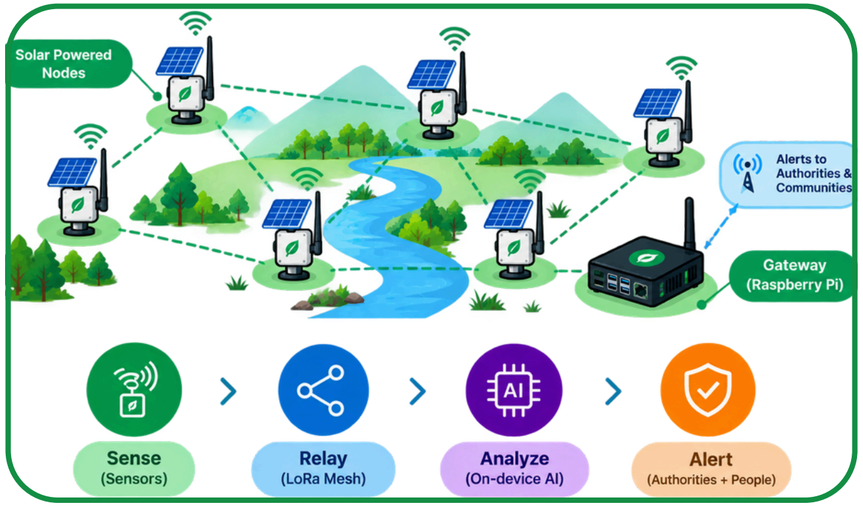


Our Solution

AegisNet 2.0 : Resilient AI-Powered Environmental Monitoring Network

A self-healing mesh of solar-powered sensor nodes that detects hazards locally, relays alerts even when towers and power grids go down, using two-tier on-device AI instead of cloud dependence.





Key Innovation

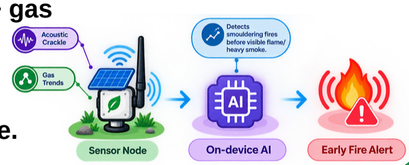
01 Self-Healing Mesh

Multi-hop LoRa routing bypasses failed nodes instead of depending on one gateway.



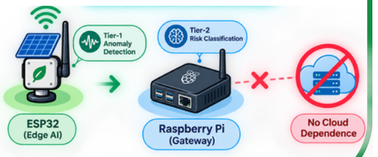
02 Pre-Flame Fire Detection

Acoustic crackle + gas trends identify smouldering fires before visible flame/heavy smoke.



03 Edge AI, Not Cloud-Dependent

Tier-1 anomaly detection runs on ESP32; Tier-2 risk classification runs at the Raspberry Pi gateway.



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TECHNICAL APPROACH

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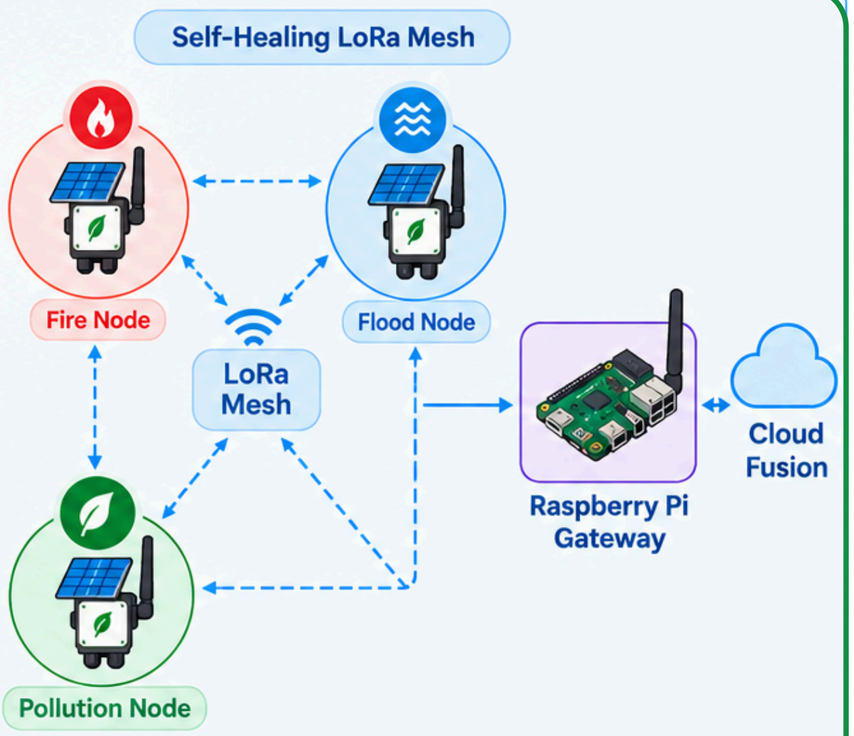
Resilient Sensing → **Edge Ai** → **Mesh** → **Risk** → **Alert**

3 Node Types

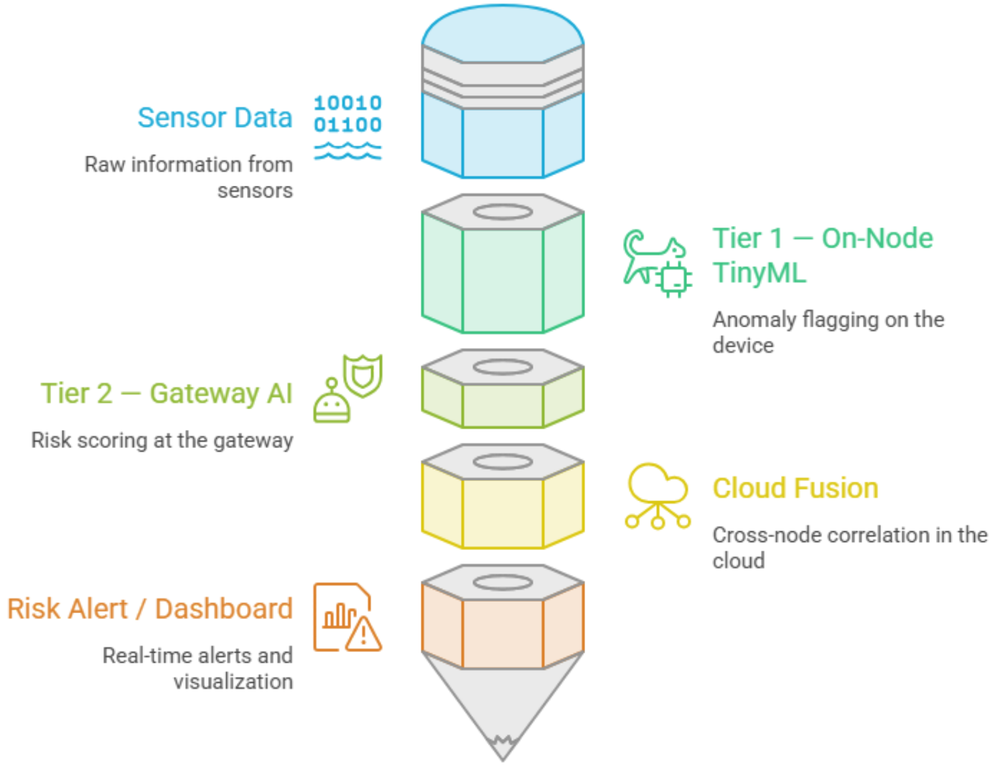
Fire Node
Acoustic + Gas + Temperature
Solar Panel → ESP32 → Sensors (acoustic/gas/temp) → LoRa SX1276/78

Flood Node
Water level + Rainfall / Environmental sensing
Solar Panel → ESP32 → Sensors (water/rain/env) → LoRa SX1276/78

Pollution Node
Air-quality / Gas sensing
Solar Panel → ESP32 → Sensors (air-quality/gas) → LoRa SX1276/78



Multi-Tier AI System Overview



ESP32 → **LoRa SX1276/78** → **Raspberry Pi** → **TFLite Micro** → **XGBoost** → **Flask / Dashboard**

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FEASIBILITY AND VIABILITY

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Why It's Feasible	Potential Challenges	HOW WE OVERCOME IT
1. Off-the-shelf Hardware 	**False Alarms** Environmental noise may trigger incorrect alerts.	**Multi-sensor Fusion + Two-tier AI** Combines multiple signals and verifies alerts across nodes.
2. Edge AI is Practical Lightweight anomaly detection can run locally on the ESP32 using TensorFlowLite Micro.	**Communication Loss** Obstructions or multiple failed nodes can disrupt routing.	**Self-healing Multi-hop LoRa Mesh** Automatically reroutes data through alternative nodes.
3. Long-range Communication LoRa provides low-power communication suitable for distributed environmental nodes.	**Power Constraints** Low sunlight can reduce node availability	**Low-power Firmware + Battery Reserve** Optimised duty cycling and energy management.
4. Off-grid Operation Solar + battery allows nodes to continue operating without grid power.	**Harsh Environment** Rain, dust and heat can affect sensors and electronics.	**IP-rated Enclosure + Calibration** Weatherproof housing and regular sensor calibration.
5. Modular Deployment New sensor nodes can be added without redesigning the entire network.	**Model Accuracy** Limited field data may reduce AI reliability.	**Continuous Field Data + Retraining** Collects real-world data and improves models over time.

Why It's Viable

- **Low Infrastructure Cost**
- **Off-grid Power**
- **Scalable**
- **Actionable Impact**

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IMPACT AND BENEFITS

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SOCIAL IMPACT

Village-Level warning,not just district-level.

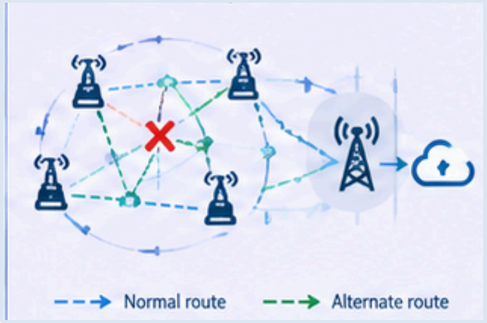
- Localiszed aleets
- (village/micro-catchment level)
- Local-language voice + buzzer
- Reaches non-smartphone users



Resilient Monitoring

Network stays alive, even when parts fail.

- Self-healing LoRa mesh
- Alternate routing after node failure
- Dual backhaul (Cellular/Wi-Fi + NB-IoT/Satellite)
- HMAC-secured packets



SCALABILITY IMPACT

Grow as per need, without rebuilding.

- Modular sensor nodes
- Incremental deployment
- Works across villages, forests & micro-catchments



ECONOMIC IMPACT

Earlier warning = lower potential loss.

- Protects crops, livestock
- & property
- Reduces disaster response cost
- Supports local livelihoods





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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

- **Smart India Hackathon 2026 – Problem Statement 26178**
  Basis for the environmental monitoring and disaster-management requirements.
- **NDMA – National Disaster Management Authority**
  Referenced as part of India's existing disaster-monitoring and response ecosystem.
- **IMD – India Meteorological Department**
  Used as the proposed source of historical rainfall archives for flood modelling.
- **CWC – Central Water Commission**
  Proposed source of historical river-gauge data for micro-catchment flood analysis.
- **Forest Survey of India (FSI)**
  Historical forest-fire incident records proposed for fire-model training.
- **CPCB – Central Pollution Control Board**
  Public AQI station data proposed for pollution-model baseline calibration.
- **ISRO – Indian Space Research Organisation**
  Referenced as part of existing satellite-based environmental monitoring systems that AegisNet complements with local sensing.

